9. If $v(x,y,z)=x^3+y^3+z^3+3xyz$, show that $\frac{\partial^2 v}{\partial y\partial z}=\frac{\partial^2 v}{\partial z\partial y}$.

10. A firm produces two types of calculators each week, x number of type A and y number of type B. The weekly revenue and cost functions (in rupees) are $R(x,y)=80x+90y+0.04xy-0.05x^2-0.05y^2$ and $C(x,y)=8x+6y+2000$ respectively.

(i) Find the profit function $P(x,y)$,

(ii) Find $\frac{\partial P}{\partial x}(1200,1800)$ and $\frac{\partial p}{\partial y}(1200,1800)$ and interpret these results.

8.6 Linear Approximation and Differential of a function of several variables

Earlier in this chapter, we have seen that linear approximation and differential of a function of one variable. Here we introduce similar ideas for functions of two variables and three variables. In general for functions of several variables these concepts can be defined similarly.

Definition 8.10

Let $A=\{(x,y)\mid a<x<b, c<y<d\}\subset\mathbb{R}^2, F:A\to\mathbb{R}$, and $(x_0,y_0)\in A$.

(i) The linear approximation of F at $(x_0,y_0)\in A$ is defined to be

$$F(x,y)=F(x_0,y_0)+\left.\frac{\partial F}{\partial x}\right|_{(x_0,y_0)}(x-x_0)+\left.\frac{\partial F}{\partial y}\right|_{(x_0,y_0)}(y-y_0) \qquad \ldots (12)$$

(ii) The differential of F is defined to be

$$dF=\frac{\partial F}{\partial x}(x,y)dx+\frac{\partial F}{\partial y}(x,y)dy, \qquad \ldots (13)$$

where $dx=\Delta x$ and $dy=\Delta y$,

Here we shall outline the linear approximations and differential for the functions of three variables. Actually, we can define linear approximations and differential for real valued function having more variables, but we restrict ourselves to only three variables.

Definition 8.11

Let $A=\{(x,y,z)\mid a<x<b, c<y<d,\ e<z<f\}\subset\mathbb{R}^3, F:A\to\mathbb{R}$ and $(x_0,y_0,z_0)\in A$.

(i) The linear approximation of F at $(x_0,y_0,z_0)\in A$ is defined to be

$$F(x,y,z)=F(x_0,y_0,z_0)+\left.\frac{\partial F}{\partial x}\right|_{(x_0,y_0,z_0)}(x-x_0)+\left.\frac{\partial F}{\partial y}\right|_{(x_0,y_0,z_0)}(y-y_0)+\left.\frac{\partial F}{\partial z}\right|_{(x_0,y_0,z_0)}(z-z_0);$$

...(14)

(ii) The differential of F is defined by

$$dF=\frac{\partial F}{\partial x}(x,y,z)dx+\frac{\partial F}{\partial y}(x,y,z)dy+\frac{\partial F}{\partial z}(x,y,z)dz, \qquad \ldots(15))$$

where $dx=\Delta x, dy=\Delta y$ and $dz=\Delta z$,

Chapter 8 Differentials and Partial Derivatives.indd 80 26-02-2020 17:02:31